

Proposal of a Singing Instruction Support System with a Residual Respiratory Volume Display in Online Lessons

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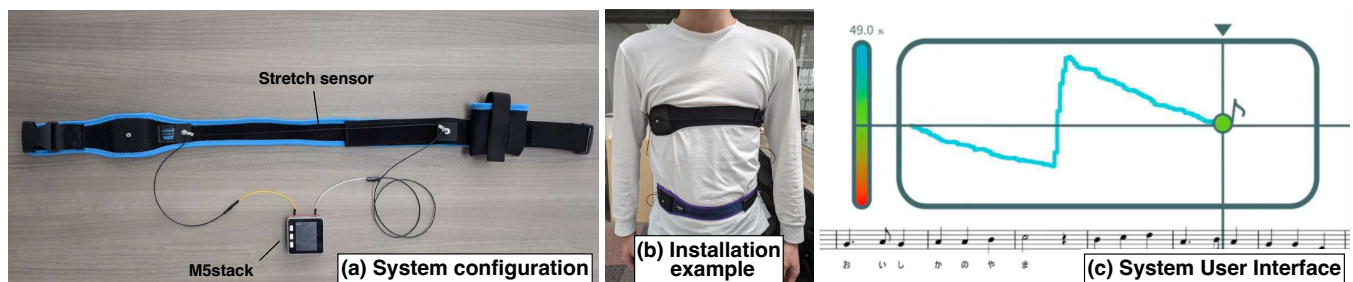


Figure 1: (a) System configuration, (b) Installation example, (c) System User Interface

Abstract

In the teaching of singing, breathing control is an important component, the observation of which has conventionally required a teacher's long experience and refined perception. However, in online lessons where visual information is limited, there is the problem that physical differences are hard to perceive and it is difficult to obtain the information necessary to provide instruction. In this research, which uses a residual respiratory volume visualization system, we verify the system's usefulness for supporting teaching. In the experiment, which imitated an online lesson, the visualization of residual respiratory volume was shown to provide fresh viewpoints for instruction.

CCS Concepts

• **Human-centered computing** → **Interaction devices; Interaction techniques.**

Keywords

Online lesson, Performance learning support, Residual respiratory volume visualization system

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1 Introduction

The promulgation of online lessons during the Covid-19 pandemic led to the verification of the effectiveness of learning and teaching in such lessons in comparison to face-to-face lessons. It has been found that online lessons have a higher degree of student satisfaction than face-to-face lessons, and have the effect of increasing autonomy and concentration [5]. In addition, it has been pointed out that, by enabling remote lessons, online lessons can potentially help solve the problems of a lack of teachers and mismatches between students and teachers [6]. From this, we consider that online lessons are not simply a replacement for face-to-face lessons, and will continue to be chosen long after the end of the pandemic.

On the other hand, when compared to face-to-face lessons, in online lessons it is difficult to obtain information needed for instruction [5]. Stabilized breathing is essential for singing, but judging whether a student's breathing process, which takes place internally, is appropriate requires minute observation and long experience. However, in an online lesson, the teacher checks the student's performance on a monitor, making it difficult to observe body movements when compared to a face-to-face lesson. Furthermore, it is



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difficult to grasp a student's actual height and chest size. Accordingly, it is also difficult to surmise what vocal volume and breathing duration a student is capable of.

The aim of this research is to construct a system to provide teaching support in online lessons by visualizing residual respiratory volume during singing. Here, residual respiratory volume refers to the amount of breath remaining when a person sings. We confer on the teacher a fresh viewpoint for instruction, by providing information about an individual student's breathing, and verify whether this contributes to teaching support.

2 Related Research

Lä et al. [3] determined that real-time visual feedback of performance information is effective at enabling more efficient mastery of singing. Several mastery support systems that display acoustic information during singing have actually been proposed [4]. However, there are few cases that focus on breathing during singing.

In general, to measure breathing during singing the following conditions must be met: (1) measurement can be carried out while the singer is standing still, (2) measurement must be carried out safely, (3) measurement equipment is small and easy to use, (4) measurement has a minimal effect on the movements involved in breathing. As a means that satisfies these requirements, there is the method of using a sensor to measure the body movements that accompany singing. In this method, the contraction of the chest or abdomen, which accompanies breathing, is measured using a device such as an inductor [2] or stretch sensor [1]. The waveform of this body movement and the waveform of breathing have been shown to have a strong relationship to the waveform measured by a flow meter [1]. In addition, a stretch sensor has the advantage of being relatively cheap. In this research, we decided to target the teacher and display residual respiratory volume to support teaching. We verify whether this is effective teaching support by presenting the residual respiratory volume to a teacher who is providing singing instruction.

3 System Design Requirements

In this research, our goal is to design and implement a teaching support system that aims, in particular, to enable mastery of breathing in singing. Specifically, this is realized by displaying residual respiratory volume, obtained from information gathered by sensors attached to the body. An example of system usage would be as a support tool in online singing instruction, in the case where the pupil is an elementary or middle school student who is still physically undeveloped to a certain extent. The determined system design requirements in this research are as follows:

The system does not restrict the user's posture or movement when singing: As the system is attached to the body, it is essential that it does not affect posture or movement during singing. In addition, ideally, anybody should be able to attach and remove the equipment easily on their own, regardless of height and chest size.
The system can estimate breathing information that enables comparison of the teacher and student : As a way for the teacher to evaluate whether the breathing presented by the system is appropriate, we can consider comparison with the breathing information of other people. Comparing the teacher and the student by the

same gauge enables the teacher to understand how the student can improve. Accordingly, it is necessary to estimate breathing information that can be compared quantitatively.

The system has a function to present breathing in real time : Observation of body movement relating to breathing is important for teaching singing, but in an online environment the restrictions of image quality and screen size make judgement difficult. Expensive equipment is required to achieve image clarity equivalent to meeting face-to-face, therefore we consider that presenting breathing in real time, in combination with a camera image, is an appropriate alternative method.

4 System Functions

4.1 Residual respiratory volume estimation

4.1.1 Breathing measurement system. The structure of the breathing measurement system and an example of wearing the device are shown in Figure 1 in (a) and (b). Stretch sensors¹ are used to measure chest movement. A stretch sensor is a variable resistor that expands and contracts like rubber. The sensors, which are fixed onto belts that can be attached and removed using buckles, are attached to two places: the chest and abdomen. The sensor values are sent to the master PC via M5Stack². M5Stack and the master PC are connected by Bluetooth.

4.1.2 Residual respiratory volume estimation. An outline of the residual respiratory volume estimation part of the system is presented in Figure 2. Reception of the data from the breathing measurement system, and the implementation of the display screen showing graphs etc., was carried out in Processing³. Estimation of residual respiratory volume was implemented in Python. The sensor data received from the breathing measurement system is sent to the estimation part via consecutive text files. As preprocessing for estimation, the sensor data offset is adjusted.

The estimation part estimates residual respiratory volume from the data received from the sensors on the chest and abdomen. The KNN (K Nearest Neighbour) method is used for estimation. The KNN method selects k learning data that are close to input data with unknown object variables, and obtains the average value of object variables of the learning data to estimate the object variable of the input data. In this research, the instantaneous sensor values from the chest and abdomen are feature values and the residual respiratory volume is the object variable.

For the k parameter, we cross validated the learning data for each experiment participant and selected the value with the smallest error. Here, error refers to the RMSE (Root Mean Squared Error). The RMSE is obtained by the following formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

- n : No. of samples
- y_i : Estimated value
- $\hat{y}_i$: Actual value

¹<https://www.imagesco.com/sensors/stretch-sensor.html> (Date of access: 2025/2/2)

²<https://docs.m5stack.com/ja/core/gray> (Date of access: 2025/2/2)

³<https://processing.org/> (Date of access: 2025/2/2)

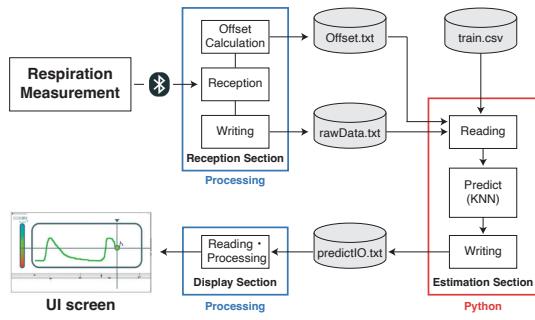


Figure 2: Overview of Residual Respiratory Volume Estimation Part

4.1.3 Dataset construction. To estimate residual respiratory volume we correlated the sensor values and residual respiratory volume and constructed datasets. A dataset is constructed each time the system user puts on the sensor belt. The user breathes into a spirometer⁴, which measures lung capacity, and the sensor values during exhalation are recorded. In addition, the fluctuating scale on the spirometer is captured with a video camera. The measurement range of the spirometer is between 1000mL and 7000mL, in degrees of 100mL. After measurement, we correlated the sensor values and residual respiratory volume for every 100mL.

4.2 Residual respiratory volume display

This function displays the system user's residual respiratory volume in real time. Figure 1 in (c) and Figure 3 provide an overview of the basic screen. The basic screen comprises a part that plots the waveform of residual respiratory volume, a part that displays residual respiratory volume by percentage and color change, and a part that displays the musical score. The waveform and the score are displayed moving across the screen from right to left, while the percentage and the color change along with the latest residual respiratory volume. This makes it possible to see the extent to which the singer is using their breath at each point on the score.

5 Verification of the Teaching Support Effect of the Residual Respiratory Volume Display Function

We verified the teaching support effect produced by visualizing the student's residual respiratory volume during singing. In a mock online lesson, we had the experiment collaborators look at a video of a student singing, and the student's residual respiratory volume, recorded beforehand. Then, we had the collaborators write down what instruction they would give the student. We analyzed how the content of the instruction changed depending on whether or not the residual respiratory volume function was used.

5.1 Experiment conditions

There were 27 experiment participants with teaching experience ranging from 0.5 to 40 years. Regarding the breakdown of teaching experience, this comprised working as a voice trainer, a teacher of

⁴<http://ttm-tokyo.com/product/kc%ef%bd%a5kcm/> (Date of access: 2025/2/2)

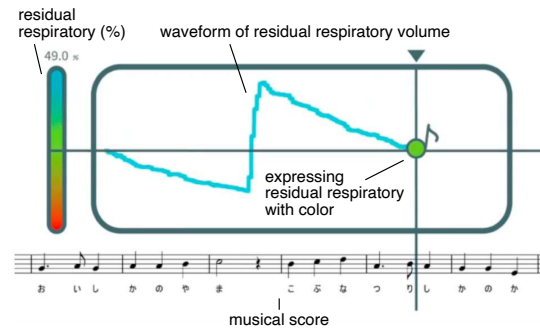


Figure 3: Screenshot of Residual Respiratory Volume Display Function

private vocal music lessons (not as preparation for music school entrance exams), a school music teacher, a school teacher from a department other than music, a choir instructor, and an accompanist. The experiment participants were recruited using the crowdsourcing platform Lancers⁵.

Using six university students acting as singing students, we recorded singing video and sensor data in advance and prepared two types of screen: one that displays only the video of a student singing, and one that displays the singing video and residual respiratory volume display function. Setting a task of aiming to produce a long tone, we asked the people acting as students to sing a long 'ah' sound.

5.2 Experiment procedure

The experiment collaborators watched the singing video of the six students and wrote down what sort of instruction they would give each student. The viewing of the video and writing of instruction content was carried out on Google Forms. To avoid ordering effects, the collaborators were divided into two groups. One group watched the video without the residual respiratory volume function first, followed by the video with residual respiratory volume function, while the other group began with the video with residual respiratory volume function, followed by the video without the function.

In the video, the same performance is repeated three times. Each performance lasts one minute. Each time they watched a video, the participants were told to write down a comment about how they would instruct the student. After the participants had watched the videos of all six students, they completed a questionnaire to evaluate the residual respiratory volume display function. The questionnaire involved giving a rating on a five-level Likert scale and writing a reason for the rating. On the Likert scale, 1 is 'Strongly disagree' and 5 is 'Strongly agree'.

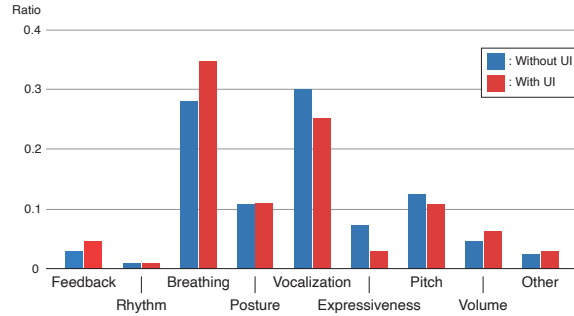
5.3 Results

5.3.1 Instruction content. We classified the instruction comments written by the experiment participants according to content and compared the comments relating to the video with the residual respiratory volume display function and the comments relating to the video without this function.

⁵<https://www.lancers.jp/> (Date of access: 2025/2/2)

Table 1: Adjusted Standardized Residuals

	Feedback	Rhythm	Breathing	Posture	Vocalization	Expressiveness	Pitch	Volume	Other
Without function	-3.42	0.14	-5.76	-0.27	4.26	7.60	2.00	-3.01	-1.00
With function	3.42	-0.14	5.76	0.27	-4.26	-7.60	-2.00	3.01	1.00


Figure 4: Proportion of Wordcount Represented by Each Category

First, all the comments were divided into single lines of text. This was because there were cases of multiple aspects being commented on for a single video. Then, the number of words in each of the divided comments was counted. The number of words in the comments about the video without the residual respiratory volume display function was 3044, while the number of words in the comments about the video with the function was 2949.

Following on, the divided comments were classified into nine categories. The categories were feedback, rhythm, breathing, posture, vocalization, expressiveness, pitch, volume, and other.

The results of calculating the proportion of the number of words represented by each category are presented in Figure 4. The breakdown is as follows: in the video without the residual respiratory volume display function, feedback accounted for 3.0% of the total wordcount, rhythm for 1.0%, posture for 10.9%, vocalization for 30.2%, expressiveness for 7.4%, pitch for 12.5%, volume for 4.7%, and 'other' for 2.5%. Regarding the video with the function, feedback accounted for 4.6%, rhythm for 1.0%, breathing for 34.8%, posture for 11.1%, vocalization for 25.2%, expressiveness for 3.0%, pitch for 10.9%, volume for 6.4%, and 'other' for 3.0%.

5.3.2 Subjective evaluation. We calculated the average values, on the five-level scale, of the objective evaluation of the system. The results were as follows. The average value was 3.74 for Q1 (This system can help realize more effective instruction), 3.56 for Q2 (Using this system enables you to think of instruction methods you've never used before), 3.59 for Q3 (Using this system gives you more confidence in the way you instruct), 3.67 for Q4 (This system helps resolve the difficulty of instruction), 3.78 for Q5 (This system helps with teaching appropriate breathing), and 3.74 for Q6 (This system helps with online singing instruction). Every rating is positive, implying the potential for visualization of residual respiratory volume to be useful for instruction.

5.4 Considerations

When a chi-squared test was carried out on the proportion of the wordcount represented by each category, a significant difference was found ($\chi^2 = 115.06$, $p < .01$). When the adjusted standardized residuals were calculated (Figure 1), it was found that the proportion was higher in the with-function video regarding breathing, feedback and volume, while the proportion was higher in the without-function video regarding expressiveness, vocalization and pitch. This demonstrates that the instruction viewpoints that the teacher focuses on change depending on whether or not the function is used.

Regarding the effect on breathing instruction in singing, the experiment showed that the system contributes to teaching support in terms of enabling a teacher to identify issues of the student's that they would not have noticed with a conventional method, and enabling objective verification of the effect of instruction. As specific examples, there were comments such as

"You can notice weak points that you haven't noticed before." (Participant 4)

"Looking at the chart you can see where breathing technique needs improving." (Participant 7)

"You can verify your own way of thinking." (Participant 12)

Regarding the effect on online lessons, it was implied that visualization of residual respiratory volume is useful for determining whether a student's breathing is appropriate.

As a specific example, we obtained comments such as

"When the movement of the chest, and so on, is difficult to see, breathing is easier to understand from the waveform presented by the system." (Participant 8)

6 Conclusion

In this research we constructed a singing support system with an inhalation volume restriction function, with the aim of supporting teaching in online singing lessons. When teachers were presented with the waveform of residual respiratory volume, and video image of students singing, the proportion of comments relating to each instruction item changed, when compared to when teachers were presented with video alone, showing that the system gives teachers different viewpoints from usual. In addition, in the subjective evaluation, the objectiveness of the system was deemed to be useful for instruction.

It is thought that, hereafter, the proposed system could be improved through being used by teachers with greater teaching experience. Furthermore, conducting an experiment in which the system is used while conversing with a student in real time would enable evaluation of whether the teaching support leads to the improvement of the student.

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